Construction and Characterization of External Cavity Diode Lasers Based on a Microelectromechanical System Device

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***Abstract*—Widely tunable external cavity diode lasers (ECDLs) with a narrow linewidth have become a powerful tool for a variety of applications. The difficulty of the present generation of ECDLs is that they are designed as laboratory instruments and are limited due to their size, high costs, and sensitivity to external influences. In this paper, we present our new ECDL design based on a microelectromechanical system (MEMS) device. This design overcomes the previous drawbacks and offers an outstanding improvement of the most important properties. Moreover, it can be adapted to any wavelength of interest and by using MEMS technology, costs can be reduced.**

***Index Terms*—Microelectromechanical devices, semiconductor lasers, laser tuning, ultrafast optics.**

I. Introduction

THE common external cavity diode lasers (ECDLs) are known for their narrow linewidth, broad tunability as well as spectral coverage and are usually based on the Littrow or Littman/Metcalf configuration [1]–[5]. Although there is a variety of ways for designing ECDLs, their further development is essential for a wide range of applications, resulting in an increasing number of optimization possibilities over time.

Therefore, important components of these ECDL implementations are the laser source with an optimal coating, the frequency-selective grating and a high reflective (HR) mirror. The semiconductor diode laser as emitting radiation source benefits from the great interest in this field of research. As a result, the fabrication of these laser types has made significant progress in recent decades. This has reduced the costs of such lasers and enables a robust and compact design [4]. The optimization of their behavior is achieved by upgrading the previously mentioned optical components. Here, the feedback of the light is controlled via the frequency-selective grating and/or the high reflective mirror, which forms the cavity with the rear facet of the laser chip. This leads to an improvement of its lasing characteristics like a narrower linewidth, a higher output power as well as greatly improved single-mode performance [1]–[4]. Because of these emission characteristics, ECDLs are increasingly gaining importance in applications fields such as healthcare, industrial process and environmental monitoring.

However, common ECDLs have their advantages and disadvantages, as they were predominantly designed as a laboratory instrument [6], [7]. Therefore, a compromise has to be made regarding their properties and the resulting costs, making them predestined for different applications.

For this purpose, we present in this paper our miniaturized ECDL design based on a micro-electro-mechanical system (MEMS) device as shown schematically in Fig. 1. This design overcomes the previous drawbacks of traditional ECDL configurations but retains their benefits. Originating from the Littman/Metcalf configuration, our modified design promises

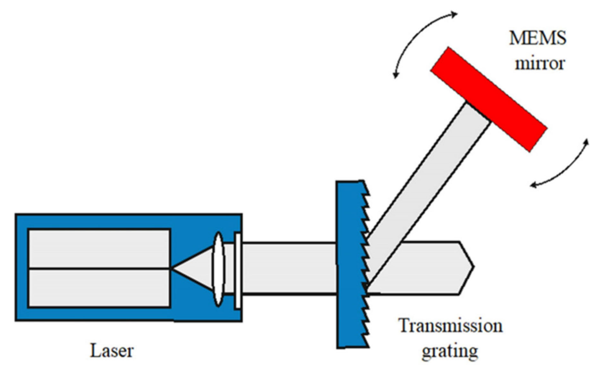

Fig. 1. The principal schematic of the MEMS based ECDL includes a laser diode, collimating lenses, a MEMS mirror, and a transmission grating.

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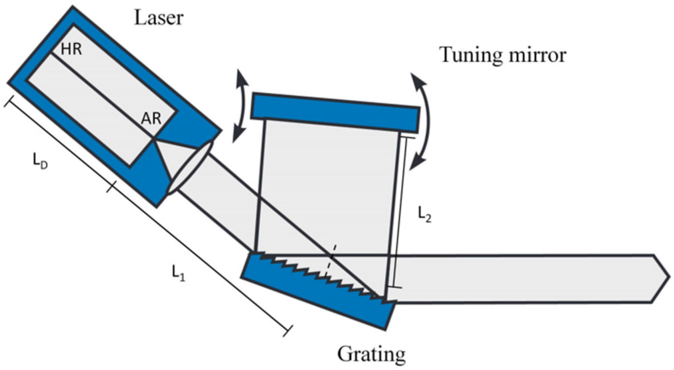

Fig. 2. The principal schematic of the classical ECDL design in Littman/Metcalf configuration. The 1st diffraction order of the grating is reflected back into the cavity. The 0th diffraction order is coupled out with the operation wavelength.

an excellent improvement in the overall system's properties. By choosing the composition of the laser chip, the system can be adapted to any wavelength or application of interest. Here we report on our first two MEMS based ECDLs that were realized with gain chips centered at 780 nm and 960 nm.

The structure of the paper is as follows. Section II presents the state of the art of the ECDLs in relation to the Littman/Metcalf configuration and explains the theoretical background of our new design. In Section III, our optimized ECDL design based on a MEMS device is shown and the most important features are illustrated. Section IV describes the use of the MEMS mirror provided by the Fraunhofer IPMS. In Section V, the results of the characterization of the developed ECDL system with a center wavelength of 780 nm is reported. This is followed by the measurement data for the ECDL system with a center wavelength of 960 nm in Section VI. In Section VII, the device-specific properties for both systems are shown. Finally, Section VIII summarizes the results and gives a conclusion and outlook on the further development of this MEM based ECDL.

II. State of the Art – Littman/Metcalf Configuration

One of the most common used designs for the present generation of ECDLs is the Littman/Metcalf configuration [1]–[5]. The schematic structure of this basic implementation is shown in Fig. 2 [8]. The frequency-selective grating and the high reflective mirror form together with the HR coated rear facet of the laser gain chip the cavity (further information see [9]). The front facet of the laser gain chip has an antireflection (AR) coating and is coupled with the external resonator. After the front facet, the laser beam first strikes the HR coated diffraction grating, which efficiency plays an essential role in relation to the chosen parameters for the structure of the cavity. This must be taken into account in order to reduce the optical losses within the ECDL system [1]. Due to the diffraction of the beam by the grating, the 1st order is reflected by the mirror back through the grating into the laser diode, the 0th diffraction order is coupled out for the respective applications with the operation wavelength. In order to suppress the 2nd order and higher ones, the correct choice of the angle of the grating is of particular importance. By tilting the mirror, the wavelength selection can be performed. This results in a change in the angle of incidence that allows the tuning of the wavelength. This relationship is described by Bragg‘s law [1]:

$$m \cdot \lambda = d \cdot (\sin\varphi_{\text{in}} + \sin\varphi_{\text{dif}}) \tag{1}$$

Where *m* is the diffraction order, λ is the operation wavelength, *d* is the grating constant, φ_{in} is the incident angle and φ_{dif} is the diffraction angle.

However, the tilt of the mirror leads not only to a change in the angle, but also causes a change in the cavity length. As a result, the frequency of the laser matches no longer the resonance of the cavity. Unless further action is taken, the modal stability of the entire system is compromised. This behavior is described by the standing wave-condition [10]:

$$\lambda = 2/N \cdot [L_D + L_1 + L_2] \tag{2}$$

Here is *N* the mode number and the term in the brackets represent the cavity length consisting of the length of the diode laser (L_D), the distance from the front facet of the laser to the diffraction (L_1) and the distance from the grating to the tuning mirror (L_2).

With the Littman/Metcalf configuration a broad mode-hop-free tuning range of the wavelength can be achieved theoretically, if it has been perfectly adapted to this behavior. In this regard, several potential approaches have been developed. These allow an adaptation of the frequency change of the feedback through a simultaneous controlled change in cavity length. One of the most widely used methods is the correct determination of the position of the pivot point around which the mirror rotates [1], [4], [11]–[13]. Through this pivot point, a continuous tuning of the wavelength can be achieved.

Although the Littman/Metcalf configuration has significantly improved the characteristics of semiconductor lasers, this design also has a few drawbacks. This is due to the fact that the current ECDLs have been designed as laboratory instruments and are sensitive to environmental disturbances such as shock, noise, and temperature fluctuations [6], [7]. An optimization of the setup and its performance is indeed possible through the choice of the used components, but was previously associated with a significant increase in costs. One way to overcome these drawbacks is the use of a MEMS device for the ECDL design. The cost-effective scalability and manufacturability of this technology is especially for the commercial development of ECDL systems of great interest and can drastically reduce their costs [14].

III. Setup of the New MEMS Based ECDL

In order to optimize the features of the current ECDL generation, a few changes have been made to the new ECDL design. The required laser chips were especially fabricated for both systems with center wavelengths at 780 nm and 960 nm. Here, the laser chips can be adapted to the required wavelength of interest by the chosen layer structure and the used materials. In

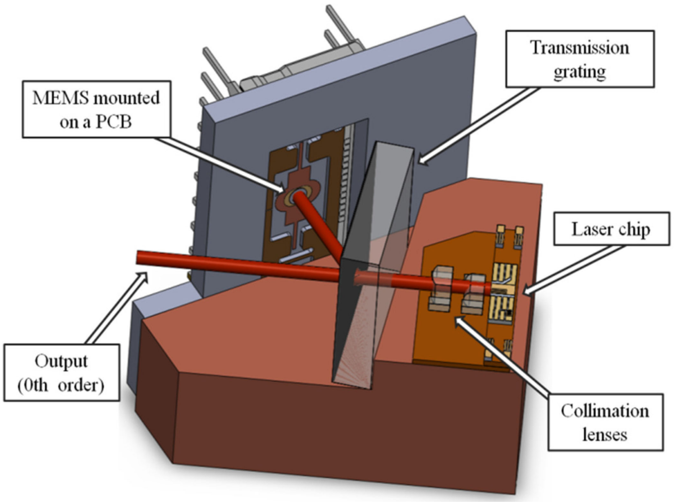

Fig. 3. CAD-Model of the MEMS based ECDL. The setup contains a semiconductor laser diode, collimating lenses, MEMS mirror mounted on a PCB and a transmission grating. The 1st diffraction order of the grating is reflected back into the cavity via the MEMS mirror. The 0th diffraction order is coupled out with the operation wavelength.

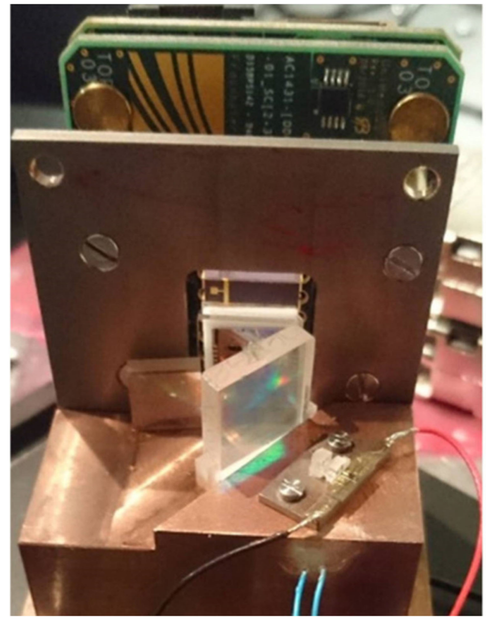

Fig. 4. Experimental setup of the ECDL and MEMS actuator mounted together on a copper chassis.

this context, GaAsP/AlGaAs as semiconductor materials were employed for the production of both gain chips to achieve the desired wavelength range. To minimize the losses of the back-reflected light of the grating into the laser diode, the front facet of the laser chip is AR coated with a reflectivity less than 0.1%. The rear facet forms together with the external optical components the cavity and is therefore HR coated with a reflectivity over 95%. To ensure good heat dissipation during operation, the laser chips are soldered on an AlN submount with collimating lenses and then mounted to a copper-tungsten micro-bench. These lenses are AR coated and have a reflectivity of less than 0.25% for both systems [7]–[15].

The major changes made to the new ECDL setup relate to the used frequency-selective grating and the high reflective mirror. Instead of a reflection grating, a transmission grating (1200 grooves/mm) is used in this setup. The transmission grating is fixed as before and has an AR coating on the backplate to avoid spurious reflections and to improve the efficiency [7]. By choosing this grating a much more compact arrangement is possible, whereby the size of the ECDL system itself is significantly reduced.

The tunability of the wavelength is then achieved by the initial use of an aluminum alloy coated MEMS mirror, which has a reflectivity in between 80% and up to 95% within the wavelength of interest. The MEMS actuator is mounted in a DIL20 housing and attached to two printed circuit boards (PCB). This construction is then mounted together with the ECDL on a copper chassis, which allows a further optimization of the temperature management [15], [16]. This assembly can be seen in Fig. 3 as a CAD-Model and in Fig. 4 as an experimental setup. As can be seen in the experimental setup, the laser was coupled out by a reflecting mirror because the PCB was slightly larger than expected.

The front facet (laser chip) is coupled to the external resonator. After the front facet, the laser beam is first collimated by cylindrical lenses with a focal length of 2.8 mm and a NA of 0.6. Subsequently, the laser beam strikes the transmission grating and is diffracted on the opposite side. In this case, the angle of the diffracted radiation depends on the respective frequency. The MEMS mirror is so arranged in this setup that the 1st diffraction order of the light is directed back to the transmission grating and from there to the laser diode. The back-reflected laser beam stabilizes the cavity of the laser at the desired operating wavelength. The output of the laser beam with the operating wavelength is via the 0th diffraction order, which is transmitted by the transmission grating. Due to the arrangement of the optical components, on the one hand, the illumination of the grating is improved and on the other hand, the power losses through the 2nd diffraction order and higher ones are reduced by optimizing the diffraction efficiency for the 0th order. Finally, the tunability of the wavelength is achieved by rotating the MEMS mirror over the static voltage of the MEMS actuator that leads to a change in the incident angle and the length of the cavity.

IV. Description of the Quasi-static 1D-MEMS Mirror

The quasi-static 1D-MEMS mirror for the ECDL is provided by the Fraunhofer IPMS. The MEMS is driven by applying a voltage to the electrostatic comb drives of the scanner. For each deflection direction, there is a set of comb drives along the torsional axis. That means that the voltage is applied to one set of comb drives for the positive direction of rotation and to the other set of comb drives for the negative direction of rotation. The achievable mechanical deflection angle goes up to $\pm 6^{\circ}$ and requires 130 V drive voltage for this scanner design. The optical active mirror plate diameter is 1.5 mm.

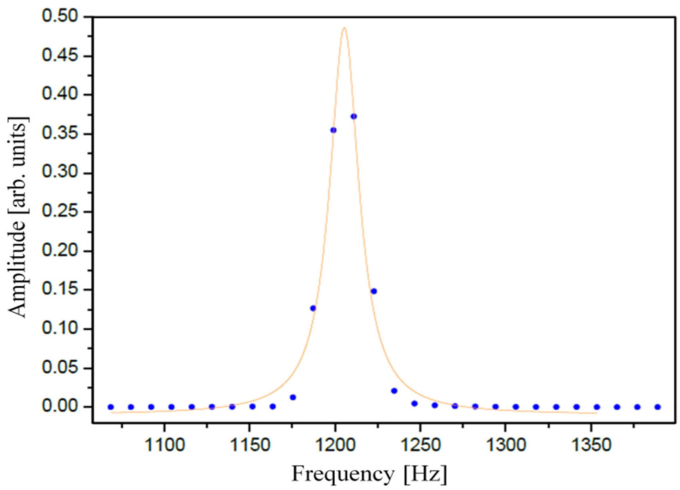

Fig. 5. Lorenz curve fit to the FFT-peak of the MEMS scanner decay curve after deflecting the scanner and turning off the drive voltage.

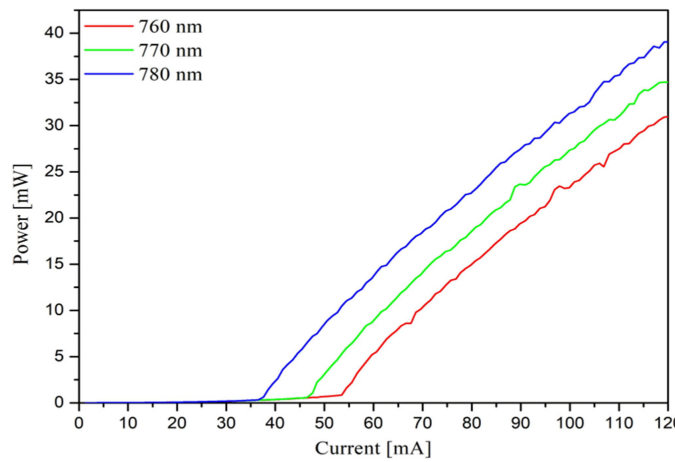

Fig. 6. PI curves for different wavelengths. For the operation wavelength at 780 nm, a power of 39 mW (at 120 mA) and a threshold current of 36.5 mA are achieved. A thermal saturation is not detectable in the measurement, which can be assumed by a higher maximum value for the output power.

A high eigenfrequency is important in order to allow a fast change between different angular positions of the mirror. Furthermore, a high eigenfrequency also enables high scan frequencies of a few hundred of hertz for scanning with linear trajectories like a saw tooth or triangular shape. The MEMS-scanner used in this experiment has an eigenfrequency of 1205 Hz, as can be seen in Fig. 5. The quasi-static movement of the scanner is realized through a staggered vertical comb drive, which is characterized by a vertical offset between the two electrostatic comb electrodes. This offset corresponds to the 75 μm BSOI-wafer device layer height used to fabricate the scanner out of single crystal silicon and is implemented by a wafer bonding process [17]. The bottom wafer comprises the mechanical design of the scanner, signal lines, bond pads and hinges that allow pushing down the electrode combs of the outer frame by stamp structures, fabricated within a top wafer. A BCB wafer bonding process activates the offset between the electrodes and keeps it in a permanent state.

In this experiment, the scanner was packaged in a standard ceramic dual inline housing DIL20 and driven in open loop. For a product setup the housing as well as the reflective coating of the mirror could be adapted specifically to the application. Moreover, the existing integrated position detection technology based on the piezo-resistive principle would be applicable.

V. MEMS Based ECDL with 780 nm Gain Chip

A. Performance

An outstanding advantage of the ECDL technology relates to its excellent lasing characteristics. For the correct selection of the laser diode for the respective applications, especially the power and the threshold current are significant characterization factors. In Fig. 6, the power as a function of the current is shown for the center emission wavelength at 780 nm of our MEMS based ECDL system together with two slightly shifted wavelengths ($\pm$10 nm). The wavelength changes are achieved by the rotation of the MEMS mirror. The current and temperature control of the laser is provided by the connected laser controller (Pilot PZ500, Sacher Lasertechnik), which enables a temperature accuracy of $\pm$0.01 °C. Unless otherwise stated, both lasers were stabilized at a temperature of 25 °C for the characterization.

For the PI curves in Fig. 6, a range of 120 mA is sampled with a step size of 0.1 mA and the respective power value is detected via a power meter (Thorlabs). In this measurement, an output power of 39 mW can be achieved for the center wavelength. Since the thermal saturation cannot be seen in any of the PI curves, higher current values can be expected to result in a further increase in the output power for the used semiconductor laser chip. The threshold current of the laser diode is 36.5 mA for the center wavelength. As expected, the threshold current shifts to higher values unless tuned at the center wavelength. This is due to the border of the gain region, since this is approximated by the tuning of the wavelength.

Of great interest is also the single-mode behavior of the diode laser, which is greatly enhanced by the use of an external cavity and is shown in Fig. 7. The spectral properties for the center wavelength of the MEMS-based ECDL can be seen there. The measurement is carried out by means of an optical spectrum analyzer (OSA) with a resolution of 0.05 nm. Here, the optical spectrum illustrates the suppression of the side modes in respect to the amplified main mode, which is represented by a high side mode suppression ratio (SMSR) of 45.5 dB and provides initial information about the stability of the system.

B. Tuning Behavior

In addition to a high output power and a good single-mode behavior of the ECDL, a broad tunability also plays a decisive role in a large number of spectroscopic methods. In our ECDL design, tuning of the wavelength is achieved by applying a voltage to the MEMS actuator. By changing the applied voltage, a tilting of the mirror is caused, which changes the angle of incidence and thus also the operating wavelength.

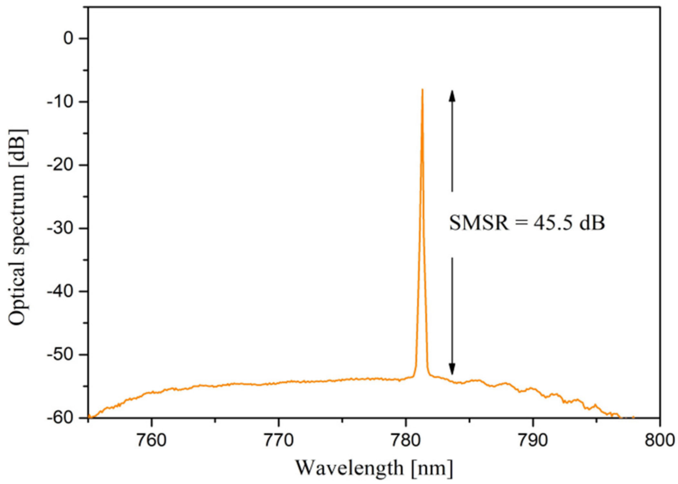

Fig. 7. The single mode behavior for the center wavelength at 780 nm is shown. The optical spectrum was recorded by an OSA over a span of about 55 nm and a resolution of 0.05 nm. A value of 45.5 dB could be determined for the SMSR.

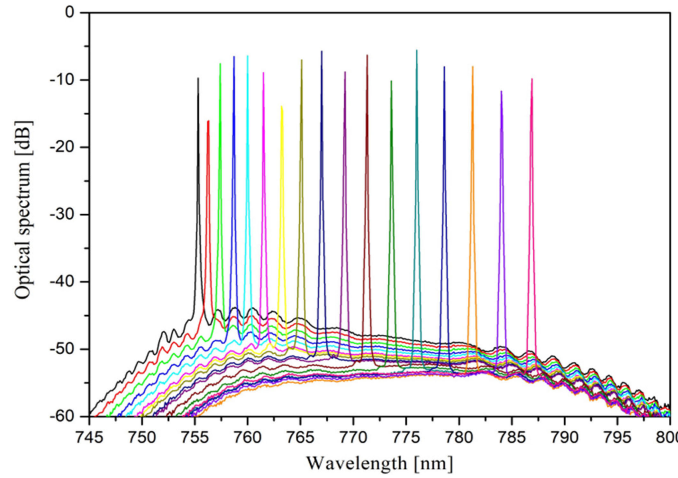

Fig. 9. Tuning behavior of the laser gain of the MEMS based ECDL. The tunability as well as the selectivity is shown around the center wavelength of 780 nm over a span of around 30 nm.

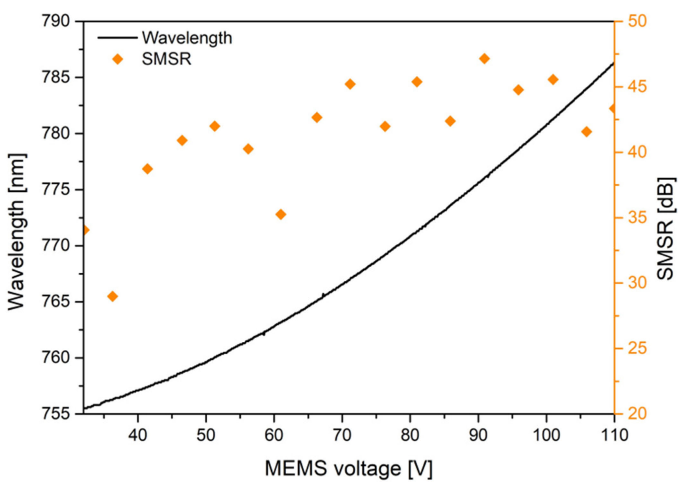

Fig. 8. Wavelength and SMSR as a function of the MEMS voltage over a range of 78 V. Changing the voltage leads to a tilting of the MEMS mirror, which in turn leads to a change in the wavelength. This allows the tuning of the wavelength over a range of about 30 nm, which is the entire available gain region of this laser chip.

This behavior of the wavelength change as a function of the applied voltage is displayed in Fig. 8. For the measurement, a range of 32 V to 110 V is tuned over the MEMS actuator and the corresponding wavelength is detected by means of a Wavemeter (Burleigh). For this system, the tuning of the wavelength can be carried out over a range of 30 nm, which is the entire available gain region of this laser chip.

Above all, the outstanding tunability of our MEMS-based ECDL should be emphasized. It allows the wavelength to be switched extremely fast by the MEMS actuator (see Section VII.B) and adjusted for the respective application. It should be kept in mind that this only serves as a crude setting. In contrast, the mode hop-free fine tuning around the operation wavelength is performed by scanning the current of the laser diode.

Also of interest is the single-mode behavior of the MEMS based ECDL, while being tuned at different wavelengths within the gain region. This is shown in Fig. 9 for multiple wavelengths over a range of about 30 nm as well as in Fig. 8 for the determined SMSRs in comparison with the associated applied voltage. The data has also been recorded with the OSA under the same measurement conditions. Although the spectral characteristics of the ECDL slightly deteriorate as it approaches the border of the gain region, there is still good suppression of the side modes and a more than adequate output power as shown in Fig. 6 for the wavelength at 760 nm. In Addition to the gain profile, the maximum tunable range depends on the used resonator configuration and can be further optimized in this regard.

VI. MEMS Based ECDL With 960 nm Gain Chip

A. *Performance*

For the evaluation of the performance of the MEMS-based ECDL with a 960 nm laser chip, the power is also detected as a function of the current for the center emission wavelength together with two slightly shifted wavelengths (±10 nm), as shown in Fig. 10. The setup is the same as described in Section IV. For the PI curves, a range of 200 mA is sampled with a step size of 0.1 mA. For the center wavelength, an output power of 65.5 mW can be achieved. In this measurement, the thermal saturation is also not visible. As a result, higher current values can also lead to a further increase in the output power for the used semiconductor laser chip. Although the PI curve for the wavelength of 940 nm and 950 nm are approximately in the same range of the output power as the center wavelength, it has the lowest threshold current of 36.5 mA due to the gain region.

The spectral characteristics for the center wavelength of the MEMS based ECDL can be seen in Fig. 11. In this case, especially the excellent suppression of the side modes is recognizable, which is reflected in a very high value for the SMSR of 85.1 dB.

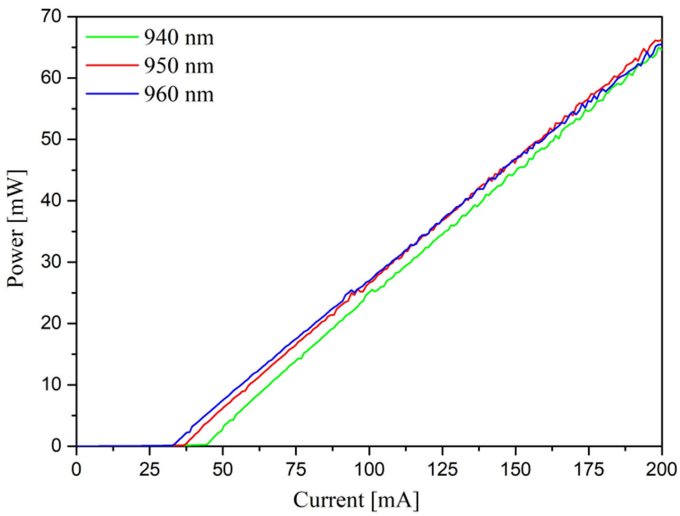

Fig. 10. PI curves for different wavelengths. For the operation wavelength at 960 nm, a power of 65.5 mW (at 200 mA) and a threshold current of 32.5 mA are achieved. A thermal saturation is not detectable in the measurement, which can be assumed by a higher maximum value for the output power.

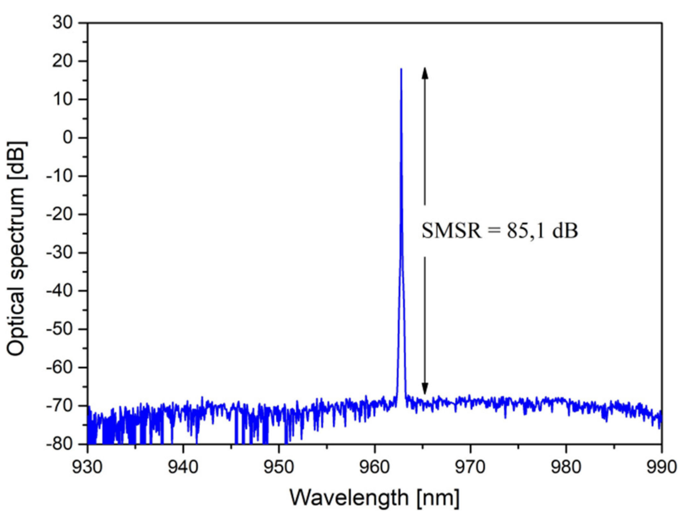

Fig. 11. The single mode behavior for the center wavelength at 960 nm is shown. The optical spectrum was recorded by an OSA over a span of about 60 nm and a resolution of 0.05 nm. An excellent value of 85.1 dB could be determined for the SMSR.

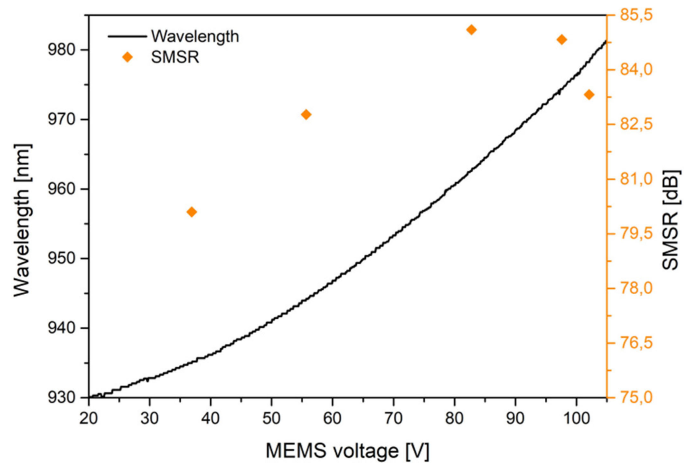

Fig. 12. Wavelength and SMSR as a function of the MEMS voltage over a range of 85 V. By changing the voltage, the wavelength can be selected. This allows the tuning of the wavelength over a range of about 50 nm, which is the entire available gain region of this laser chip.

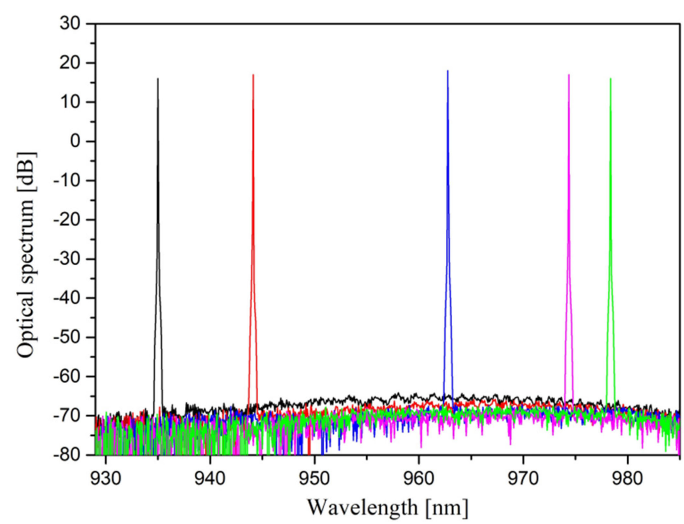

Fig. 13. Tuning behavior of the laser gain of the MEMS based ECDL. The tunability as well as the selectivity is shown around the center wavelength of 960 nm over a span of around 45 nm.

B. Tuning Behavior

For the detection of the maximum tuning range, the wavelength is recorded as a function of the applied voltage at the MEMS actuator and is demonstrated in Fig. 12. For the measurement, a range of 20 V to 105 V is sampled via the MEMS actuator, which corresponds to a wavelength change of 50 nm in total and represents the entire available gain region of this laser chip.

Again, the wide tuning range only serves to set the desired wavelength for the respective application extremely fast. The mode hop-free fine tuning around the operation wavelength is performed by scanning the current of the laser diode as before.

Fig. 13 also shows the single-mode behavior of the ECDL while being tuned at different wavelengths within the gain region. It can be seen here that the spectral characteristics are maintained at a high level during the wavelength tuning, which is reflected in a high SMSR. The determined SMSRs from this data can be seen in Fig. 12 in comparison with the associated applied voltage.

VII. Device-Specific Factors for the MEMS Based ECDL Systems

The repeatability, switching speed, relative intensity noise (RIN) and static mode stability are device-specific factors and extremely important for a variety of applications in order to assess the significance of the data from the measurements [18], [19]. Since both developed MEMS based ECDLs consist of the

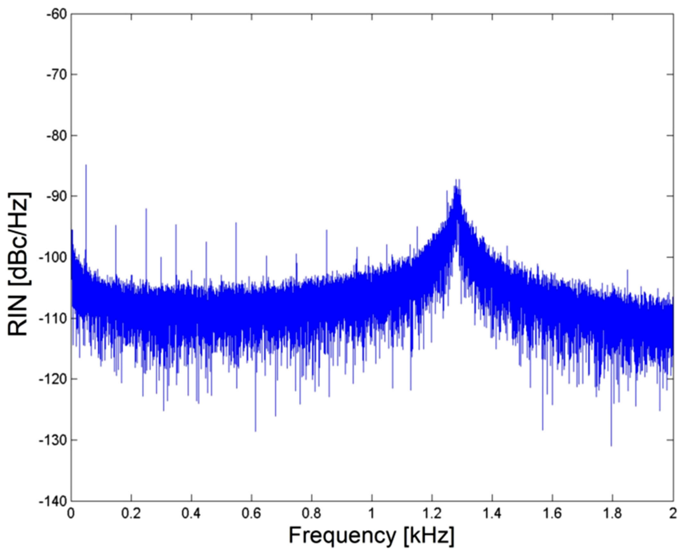

Fig. 14. The RIN measurement of the system is shown for a span of 2 kHz. The resonance frequency of the MEMS actuator could be clearly determined by 1.3 kHz. The noise floor offers a very low value of −110 dBc/Hz.

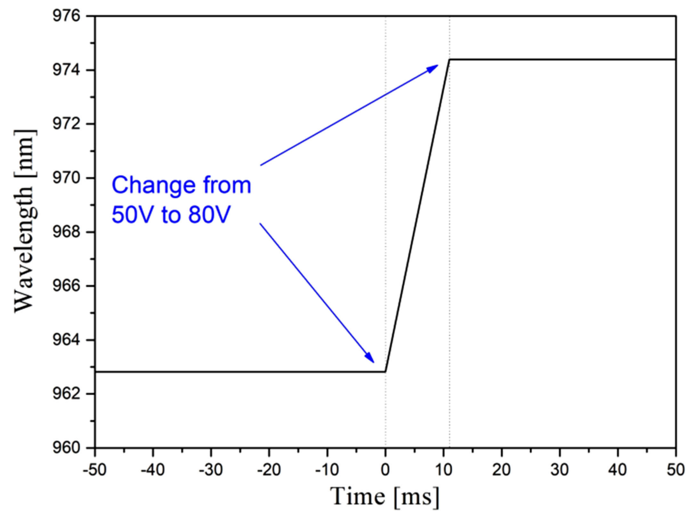

Fig. 15. The measurement of the switching speed between 2 wavelengths takes place by means of a voltage change in the MEMS actuator. A change from 50 V to 80 V corresponds to a wavelength change of about 11.5 nm, which could be measured with a switching time of 11 ms.

same basic components and therefore similar results to these factors are measured, only results for one ECDL are displayed.

A. Relative Intensity Noise

An extremely important factor is the RIN of the MEMS actuator, which provides information about the stability of the intensity and can be a limiting factor for various applications. This is due to the fact that fluctuations in the optical output power reduce the signal-to-noise ratio (SNR), resulting in deterioration of the detection limit of the entire system [8].

The measurement was performed by stabilizing the temperature of the MEMS based ECDL using the PilotPZ500 laser controller. Subsequently, the output of the system has been detected by means of a photodiode (Thorlabs). The photodiode is also connected to a digitizer (Agilent L4532A), which records the voltage at the analog output of the photodiode for a period of 10 s at a sampling rate of 1 MSa/s and can be seen in Fig. 14.

In this case, a value of 1.3 kHz could be determined from the power spectral density (PSD) for the resonant frequency of the MEMS actuator. It should be noted that this value is one of the main limiting factors for the switching speed of the MEMS actuator [16], [20]. Due to the excellent results, an improvement of the switching speed is expected. In addition, the overall noise floor offers a very low value of −110 dBc/Hz.

B. Switching Speed

An advantage of the MEMS technology is that the used MEMS actuator for the tuning of the wavelength provides a resonant frequency in the kHz range, which greatly reduces the switching speed of the system [8]. In order to determine the switching speed, the static voltage on the MEMS actuator is changed from 50 V to 80 V. This corresponds to a wavelength change of $\Delta\lambda \approx 11.5$ nm, which has been measured with a switching time of 11 ms and is shown in Fig. 15. The measurement shows that the achieved switching speed allows a fast change between the wavelengths during a defined change in the applied voltage.

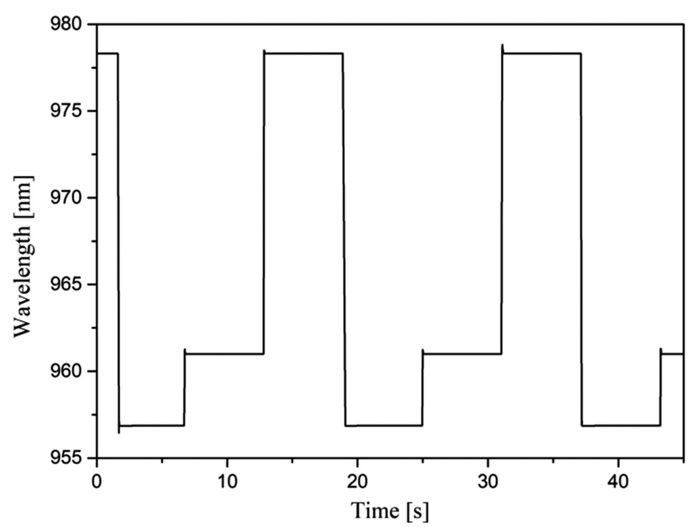

Fig. 16. The repeatability of the measurement over a period of 45 s when the wavelength is changed by pre-determined voltage values and then reproduced.

C. Repeatability

For repeatability, previously defined voltages are changed in a short-term measurement over a period of 45 s and then reproduced as shown in Fig. 16. A major advantage over the classical configurations in this area is the use of the MEMS technology. This makes it possible to set the desired wavelength precisely via the predefined voltages of the MEMS actuator, without further adjustments being necessary [8].

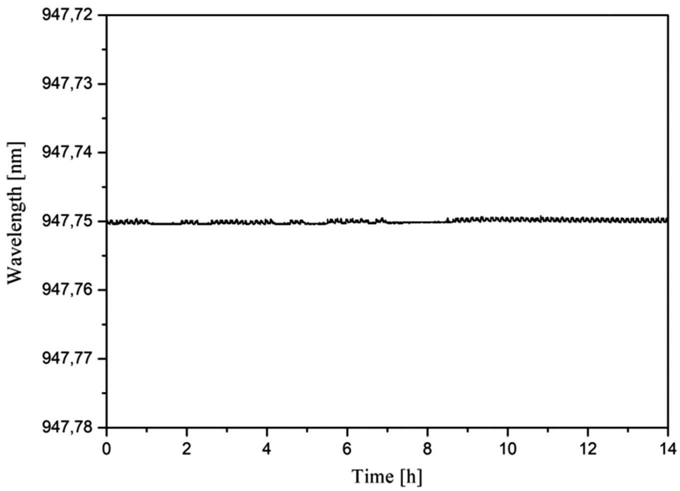

Fig. 17. Long-term measurement of the MEMS-based laser system over a period of 14 h with a maximum drift of 0.8 pm.

D. Static Mode Stability During Long-Term Measurement

Another important and decisive factor of such a system is the static mode stability during a measurement. As already mentioned in Section II, the previous ECDLs have some drawbacks, as they have been constructed as a laboratory instrument. This relates above all to the sensitivity to environmental influences, which can adversely affect the measurement conditions and is a critical factor for a variety of applications. For this purpose, a measurement is carried out at a pre-determined wavelength over a period of 14 h and can be seen in Fig. 17.

For the identification of the wavelength drift during this period, the highest deviation was determined from the data set. Due to the excellent mechanical design as well as the thermal management of the MEMS-based ECDL system, a maximum wavelength drift of only 0.8 pm is measured. This measurement shows that the ECDL can operate at the required wavelength within this period of time without the need for a readjustment.

VIII. Conclusion

In this paper we have introduced our new ECDL design in combination with the promising MEMS technology. Two systems were constructed and characterized which operate around a center emission wavelength of 780 nm and 960 nm. The use of a MEMS mirror in conjunction with a transmission grating not only allows a more compact and cost-efficient design, but also the reduction of the mechanical components. This is reflected in the excellent results in terms of repeatability and static mode stability of the system and is therefore of great interest for applications such as Raman spectroscopy, in which a high level of precision is in the focus. In addition, it provides excellent performance and is tunable over a broad wavelength range. It has been shown that the new MEMS based ECDL design offers many benefits over the classic configuration, but also has further optimization potential, e.g., the optimization of the influence of the grating and the MEMS structures.

For this purpose, an innovative ECDL design has already been developed in our company, which also incorporates the MEMS technology. This provides an even more compact arrangement and allows the correction of the cavity length during the wavelength tuning. In addition, this design employs a reflection grating instead of a transmission grating and prospects together with the MEMS mirror a much better immunity to mechanical stress as well as a higher tuning speed of the wavelength. Such a system is already in the test phase for an InP-based laser chip with a center emission wavelength of 1550 nm and the first results will be presented as soon as possible.

A MEMS based external cavity interband cascade laser is also currently under development. The advantage of the ICL technology is that the mid-infrared range can also be covered. This area involves stronger absorption lines of significant gases and is of great interest to a variety of applications.

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